

INSTALLATION GUIDE

App version: v0.1.0-dev

This guide must be updated after installation, infrastructure, dependency, model, database, backup or environment changes.

OVERVIEW

This application runs on an Ubuntu 24.04 computer with OpenClaw, PostgreSQL, Ollama qwen3:4b, FastAPI, React and background workers.

REQUIRED SERVICES

- OpenClaw
- PostgreSQL
- Ollama with qwen3:4b
- FastAPI backend
- React frontend
- Background workers

ENVIRONMENT VARIABLES

Use .env.example as the template. Do not commit the real .env file.

POSTGRES SQL SETUP

Use Docker Compose by default because the repository already includes a local-only PostgreSQL service. Native PostgreSQL is acceptable later if it is configured with the same local-only exposure and environment-variable credential rules.

PostgreSQL must not be exposed to the internet. The included Compose configuration binds PostgreSQL to localhost only:

```
text
127.0.0.1:5432:5432
```

Create the real .env from .env.example, then replace placeholder database credentials with local secrets. Do not commit .env.

```
bash
cp .env.example .env
EDIT POSTGRES_PASSWORD AND DATABASE_URL IN .ENV ONLY
```

Start PostgreSQL with Docker Compose:

```
bash
docker compose up -d postgres
```

Apply Stage 0 migrations:

```
bash
./scripts/db_migrate.sh
./scripts/db_status.sh
```

The migration runner loads .env when present, creates schema_migrations, and applies SQL files from database/migrations in filename order. The first migration enables required extensions and creates the Stage 0 model-task logging table.

Required extensions for the first migration:

- pgcrypto
- uuid-ossp

Stage 0 tables:

- schema_migrations
- model_task_logs

OLLAMA SETUP

Ollama must run locally only and the approved V1 local model is qwen3:4b.

Recommended secure defaults for a user-space service:

```
bash
mkdir -p ~/.config/systemd/user
cat > ~/.config/systemd/user/ollama.service <<'EOF'
[Unit]
Description=Ollama local model server (user-space)
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
Environment=OLLAMA_HOST=127.0.0.1:11434
Environment=OLLAMA_MODELS=/home/robot/.ollama/models
Environment=PATH=/home/robot/.local/bin:/usr/local/bin:/usr/bin:/bin
ExecStart=/home/robot/.local/bin/ollama serve
Restart=on-failure
RestartSec=5
```

```
[Install]
WantedBy=default.target
EOF
```

```
systemctl --user daemon-reload
systemctl --user enable --now ollama.service
```

The host should have user lingering enabled so the service starts after reboot. Validate with:

```
bash
loginctl show-user "$USER" -p Linger
systemctl --user is-enabled ollama.service
systemctl --user is-active ollama.service
```

Install and validate the approved model:

```
bash
./scripts/install_ollama_qwen3_4b.sh
./scripts/test_ollama_qwen3_4b.sh
```

The validation scripts fail if Ollama is unreachable, if qwen3:4b is missing, or if unexpected local Ollama models are present.

PAID MODEL SETUP

Configure ChatGPT as the primary development model and Anthropic as fallback.

Stage 0 model routing policy:

1. Ollama qwen3:4b for low-risk cleanup, classification, extraction, tagging and first-pass summaries.
2. ChatGPT/OpenAI as the primary paid development/reasoning provider.
3. Anthropic as fallback when ChatGPT/OpenAI token, context, rate or quota limits block progress.
4. Latest suitable paid models only for final higher-risk production/research analysis later.

Configure API keys only in the real .env file or host secret manager:

```
text
OPENAI_API_KEY=...
ANTHROPIC_API_KEY=...
```

Do not commit real API keys. Do not expose paid provider keys to frontend code.

Fallback events are recorded through the `model_task_logs` foundation. Logs must not store prompts, raw responses, API keys or secrets.

BACKUP AND RESTORE

Use the backup and restore scripts in `/backup`.

Stage 0 scripts:

- `backup/backup_postgres.sh`
- `backup/restore_postgres.sh`
- `backup/backup_app_config.sh`
- `backup/restore_app_config.sh`

PostgreSQL backup and restore scripts load `.env` when present. Backups use custom-format `pg_dump` files and write a `.sha256` checksum. Restore verifies the checksum when the checksum file is present.

Do not include unencrypted secrets in backups.

VALIDATION CHECKLIST

Run structural validation:

```
bash
./scripts/stage0_validate.sh
```

Stage 0 checklist:

- Backend skeleton imports and exposes `/health`
- Frontend skeleton exists and builds once Node dependencies are installed
- PostgreSQL Docker Compose config binds to localhost only
- Migration runner creates `schema_migrations` and Stage 0 migration creates `model_task_logs`
- Ollama is installed locally on `127.0.0.1:11434` and responds with `qwen3:4b`
- ChatGPT primary model is configured through `environment/provider` settings
- Anthropic fallback is configured through `environment/provider` settings
- Mock fallback tests cover token/rate-limit fallback behaviour
- `model_task_logs` migration foundation exists for fallback event logging
- ASVS register exists and records that latest stable ASVS must be checked at audit time
- No secrets are committed